

Pure Kolmogorov-Arnold Network for Single-Molecule Localization Microscopy

Technical Implementation Report

February 2026

1. Introduction

This report presents a pure Kolmogorov-Arnold Network (KAN) implementation designed for sub-pixel emitter localization in Single-Molecule Localization Microscopy (SMLM). The network predicts precise nanometer-scale coordinates of fluorescent emitters from 7×7 pixel Point Spread Function (PSF) patches. We describe implementations for single-emitter, two-emitter, and three-emitter scenarios.

2. Mathematical Foundation

2.1 Kolmogorov-Arnold Representation Theorem

The Kolmogorov-Arnold representation theorem states that any multivariate continuous function can be represented as a composition of univariate functions and addition. For a function $f: [0,1]^n \rightarrow \mathbb{R}$:

$$f(x_1, \dots, x_n) = \sum_{\gamma} \Phi_{\gamma} \left(\sum_i \varphi_{\gamma,i}(x_i) \right)$$

where $\varphi_{\gamma,i}$ and Φ_{γ} are continuous univariate functions.

2.2 Pure KAN Layer Formulation

Our pure KAN layer computes each output as a sum of learnable univariate functions applied to each input:

$$y_o = \sum_i \varphi_{o,i}(x_i)$$

Each univariate function $\varphi_{o,i}$ is parametrized as a B-spline:

$$\varphi_{o,i}(x) = \sum_{\ell} w_{o,i,\ell} \cdot B_{\ell}(x)$$

where $B_{\ell}(x)$ are B-spline basis functions and $w_{o,i,\ell}$ are learnable weights. This formulation maintains mathematical purity—no residual connections, no base activation functions, only learnable splines.

3. Network Architecture

3.1 Architecture Overview

The network consists of 5 stacked pure KAN layers that progressively reduce dimensionality from the flattened input to the output coordinates. The architecture varies only in the final output dimension based on the number of emitters:

Layer	1 Emitter	2 Emitters	3 Emitters
Input	49	49	49
Layer 1	49 → 64	49 → 64	49 → 64
Layer 2	64 → 32	64 → 32	64 → 32
Layer 3	32 → 16	32 → 16	32 → 16
Layer 4	16 → 8	16 → 8	16 → 8
Layer 5 (Output)	8 → 2	8 → 4	8 → 6
Output Format	(x, y)	(x ₁ ,y ₁ ,x ₂ ,y ₂)	(x ₁ ,y ₁ ,x ₂ ,y ₂ ,x ₃ ,y ₃)

Table 1: Network architecture comparison across single, dual, and triple emitter configurations.

4. B-Spline Activation Functions

4.1 B-Spline Basis Functions

B-splines (Basis splines) are piecewise polynomial functions defined by a knot vector. We use cubic B-splines (order $k=3$) which provide smooth, twice continuously differentiable activation functions. The B-spline basis functions are computed recursively using the Cox-de Boor formula.

4.2 Grid Configuration

Parameter	Value	Description
Grid Size	12	Number of intervals in the grid
Spline Order (k)	3 (cubic)	Polynomial degree of splines
Number of Basis Functions	15	$\text{grid_size} + \text{spline_order} = 12 + 3$
Grid Range (Layer 1)	[-0.1, 1.1]	Matches normalized input range [0,1]
Grid Range (Layers 2-5)	[-3.0, 3.0]	Wider range for intermediate values
Extended Knots	19	$\text{grid_size} + 2k + 1 = 12 + 6 + 1$

Table 2: B-spline grid configuration parameters.

4.3 Grid Range Design Rationale

The grid range is critical for effective spline learning. We use different ranges for different layers:

- **Layer 1 ([−0.1, 1.1]):** The input is normalized to approximately [0, 1]. We extend slightly beyond to handle outliers without information loss. This ensures all basis functions are utilized (unlike the original [−1, 1] range where half would be wasted on negative values that never occur in pixel intensities).
- **Layers 2-5 ([−3.0, 3.0]):** Intermediate activations can take larger positive or negative values. The wider range accommodates the output distribution of previous layers while maintaining spline coverage.

5. Input and Output Specification

5.1 Input Data

The input consists of 7×7 pixel patches extracted around detected PSF centers from SMLM images. Each patch captures the intensity distribution of fluorescent emitter(s).

Property	Specification
Raw Shape	(N, 7, 7) — N samples, 7×7 pixels each
Flattened Shape	(N, 49) — 49 input features per sample
Data Type	Float32 (raw pixel intensities, non-negative)
Typical Range	[0, ~10000] photon counts (camera-dependent)
File Format	TIFF stack (train_frames.tif, val_frames.tif, test_frames.tif)

5.2 Output Data

The output represents predicted emitter coordinates in nanometers. The network outputs standardized values which are then unstandardized to recover physical coordinates.

Configuration	Output Dim	Format	Units
1 Emitter	2	[x, y]	nanometers
2 Emitters	4	[x ₁ , y ₁ , x ₂ , y ₂]	nanometers
3 Emitters	6	[x ₁ , y ₁ , x ₂ , y ₂ , x ₃ , y ₃]	nanometers

Table 4: Output specifications for each emitter configuration.

6. Normalization Strategy

6.1 Input Normalization

Raw pixel intensities are normalized using a simple scaling approach:

$$x_{\text{normalized}} = x_{\text{raw}} / x_{\text{scale}}$$

where x_{scale} is the 99.5th percentile of training data pixel values. This approach:

- Maps most values to [0, 1] while allowing outliers to exceed 1 slightly
- Preserves information (no clamping that would destroy bright pixel data)
- Aligns with first layer's grid range [-0.1, 1.1] ensuring full basis function utilization
- Uses training statistics only (computed once, applied to train/val/test)

6.2 Target Standardization

Output coordinates are standardized (z-score normalization) during training:

$$y_{\text{standardized}} = (y_{\text{nm}} - y_{\text{mean}}) / y_{\text{std}}$$

where y_{mean} and y_{std} are computed from training coordinates. This is reversed during inference:

$$y_{\text{nm}} = y_{\text{predicted}} \times y_{\text{std}} + y_{\text{mean}}$$

6.3 Rationale for Target Standardization

Standardization is essential for several reasons:

1. **Numerical Stability:** Raw coordinates (e.g., 3500 nm) would require very large network weights. Small standardized values (~ 0) are easier for the network to learn.
2. **Balanced Learning:** Without standardization, if x-coordinates cluster around 3500 nm and y-coordinates around 100 nm, the MSE loss would prioritize x accuracy. Standardization equalizes the importance of both coordinates.
3. **Interpretable Loss:** MSE on standardized targets measures error in standard deviations, making training progress comparable across different datasets.

7. Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	1e-3 (0.001)
Epochs	100

Batch Size (Training)	128
Batch Size (Validation/Test)	256
Loss Function	Mean Squared Error (MSE)
Gradient Clipping	Max norm = 1.0
Weight Initialization	Gaussian ($\mu=0$, $\sigma=0.1$)
Model Selection	Best validation MSE checkpoint

Table 5: Training hyperparameters and configuration.

8. Multi-Emitter Handling

8.1 The Permutation Ambiguity Problem

For multi-emitter configurations, there is an inherent ambiguity: if a frame contains emitters at positions A and B, the ground truth could be stored as [A, B] or [B, A]. A naive MSE loss would penalize correct predictions that happen to be in the "wrong" order.

8.2 Solution: Sorting by X-Coordinate

We resolve this by establishing a canonical ordering: emitters are always sorted by their x-coordinate. This is applied to both ground truth labels (during data loading) and network predictions (during training and inference). The sorting operation:

- Is applied as a post-processing step after the network forward pass
- Does not affect the KAN architecture's mathematical purity
- Is differentiable (gradients flow through the reordering)
- Adds no learnable parameters

9. Complete Data Pipeline

The complete data flow from raw input to final output is summarized below:

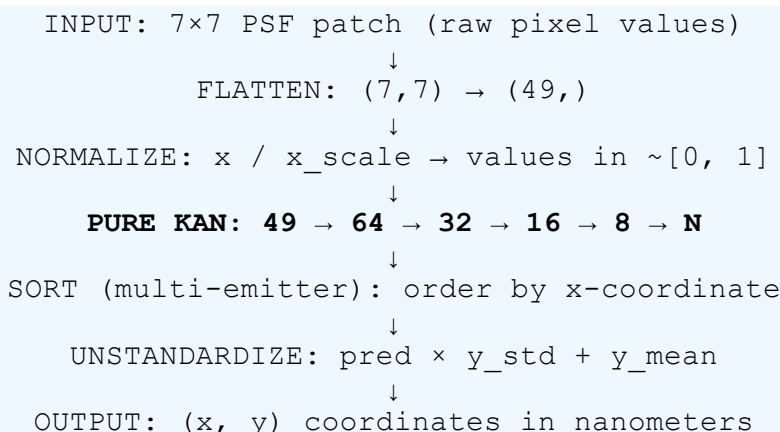


Figure 1: Complete data pipeline from raw PSF patch to predicted emitter coordinates.

10. Evaluation Metrics

Model performance is evaluated using standard localization metrics computed on the test set:

- **RMSE (Root Mean Square Error):** $\sqrt{\text{mean}(\text{error}^2)}$ — measures average localization precision
- **MAE (Mean Absolute Error):** $\text{mean}(|\text{error}|)$ — robust to outliers
- **Per-emitter Error:** $\sqrt{(x_pred - x_true)^2 + (y_pred - y_true)^2}$ — Euclidean distance per emitter

11. Summary

This report presents a mathematically pure Kolmogorov-Arnold Network implementation for SMLM emitter localization. Key design decisions include: (1) pure B-spline layers without residual connections, (2) layer-specific grid ranges aligned with expected value distributions, (3) simple input normalization preserving information, (4) target standardization for stable training, and (5) sorting-based multi-emitter handling. The architecture scales naturally from single to multiple emitters by adjusting only the output dimension while maintaining identical preprocessing and training procedures.